

Groundwork

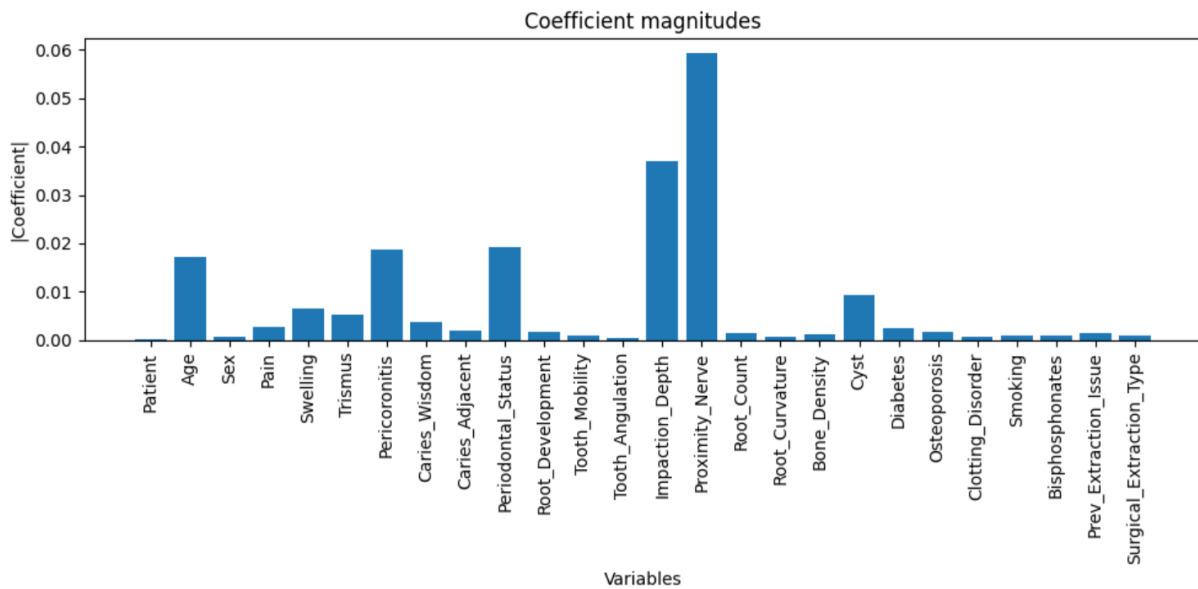
We now take a step back and investigate some simpler models alongside the MLP approach. Although the MLP model achieved strong predictive performance, simpler models are often preferred when they provide sufficient accuracy, since they are easier to interpret and explain. In particular, linear regression models allow direct inspection of the learned coefficients, making it possible to identify which variables are associated with increased or decreased complication risk. In our groundwork we use the dataset generated by Jonas, who worked on our project previously. The dataset contains 50000 data points where 3000 data points is used as test set. The work can be found in the "Groundwork" folder.

Binary Target

We begin by fitting a multiple linear regression model to investigate the relationship between the input features and a new binary target variable where a value of 1 indicates that a patient experienced one or more complications, while 0 indicates no complications.

Before training the model, the input features were standardized to ensure that the regression coefficients are directly comparable across variables.

The figure below shows the estimated regression coefficients for the model.

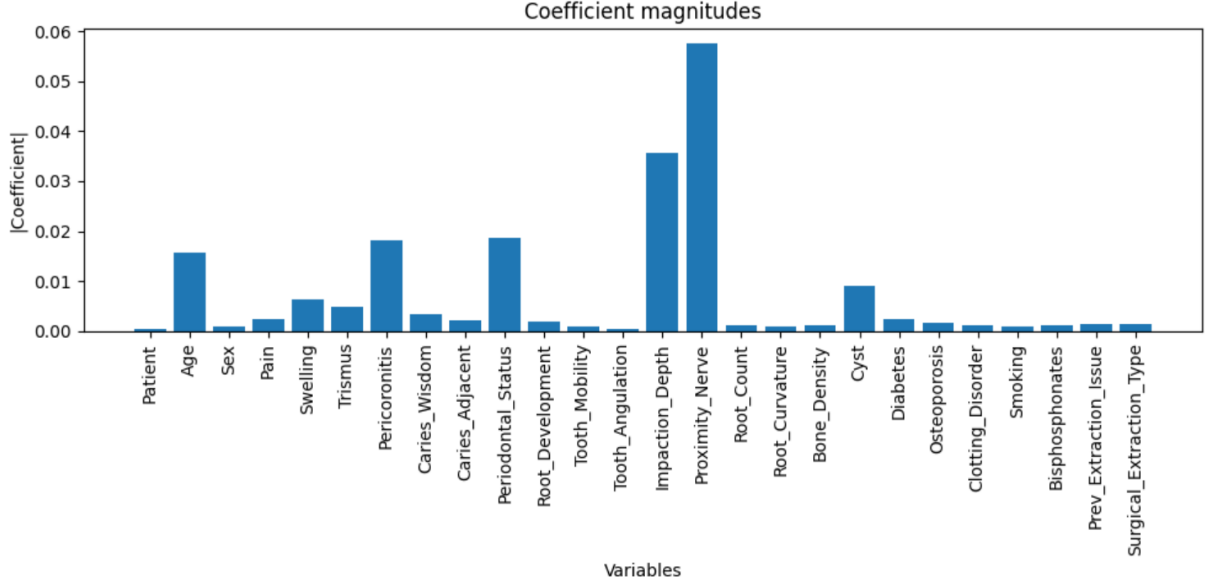


The regression coefficients indicate that variables such as proximity to the nerve and impaction depth have the strongest influence on the predicted complication risk.

We now extend the analysis to a federated setting. Instead of training a single regression model on the full dataset, a separate multiple linear regression model is trained locally on each client using the same binary target variable.

After local training, the regression coefficients are aggregated using a weighted average based on the number of samples at each client. Which produces a global federated model.

The figure below shows the estimated regression coefficients for the model.



We see that the coefficients of the federated multibell regression are close to those of the centralized setup.

We also do all these steps using logistic regression which yields a similar plot. However, we later choose to drop this approach.

Four models

We now extend the analysis by training one regression model for each of the four complication types.

The overall method remains the same in both the centralized and federated settings.

Since the models now produce separate predictions for each complication, the predicted risk scores can be categorized into the classes low, medium, and high using percentile threshold. This makes it possible to evaluate the models using the F1-macro score across the three risk categories.

```

--- Risk_AlveolarOsteitis ---
True column: Risk_Category_AlveolarOsteitis
Pred column: Risk_AlveolarOsteitis_group
Class 0: Precision=0.765, Recall=0.793, F1=0.779
Class 1: Precision=0.545, Recall=0.531, F1=0.538
Class 2: Precision=0.715, Recall=0.709, F1=0.712
Macro F1 = 0.676

--- Risk_SecondaryInfection ---
True column: Risk_Category_SecondaryInfection
Pred column: Risk_SecondaryInfection_group
Class 0: Precision=0.702, Recall=0.732, F1=0.717
Class 1: Precision=0.550, Recall=0.529, F1=0.539
Class 2: Precision=0.747, Recall=0.746, F1=0.746
Macro F1 = 0.667

--- Risk_NerveDysesthesia ---
True column: Risk_Category_NerveDysesthesia
Pred column: Risk_NerveDysesthesia_group
Class 0: Precision=0.745, Recall=0.813, F1=0.778
Class 1: Precision=0.607, Recall=0.553, F1=0.579
Class 2: Precision=0.746, Recall=0.757, F1=0.751
Macro F1 = 0.703

--- Risk_Bleeding ---
True column: Risk_Category_Bleeding
Pred column: Risk_Bleeding_group
Class 0: Precision=0.549, Recall=0.640, F1=0.591
Class 1: Precision=0.376, Recall=0.403, F1=0.389
Class 2: Precision=0.642, Recall=0.531, F1=0.582
Macro F1 = 0.520

=====
CENTRALIZED FINAL F1 = 0.642
=====

```

(a) Centralized

```

--- Risk_AlveolarOsteitis ---
True column: Risk_Category_AlveolarOsteitis
Pred column: Risk_AlveolarOsteitis_group
Class 0: Precision=0.765, Recall=0.793, F1=0.779
Class 1: Precision=0.522, Recall=0.508, F1=0.515
Class 2: Precision=0.698, Recall=0.692, F1=0.695
Macro F1 = 0.663

--- Risk_SecondaryInfection ---
True column: Risk_Category_SecondaryInfection
Pred column: Risk_SecondaryInfection_group
Class 0: Precision=0.686, Recall=0.715, F1=0.700
Class 1: Precision=0.538, Recall=0.518, F1=0.528
Class 2: Precision=0.733, Recall=0.732, F1=0.732
Macro F1 = 0.653

--- Risk_NerveDysesthesia ---
True column: Risk_Category_NerveDysesthesia
Pred column: Risk_NerveDysesthesia_group
Class 0: Precision=0.742, Recall=0.810, F1=0.775
Class 1: Precision=0.621, Recall=0.566, F1=0.592
Class 2: Precision=0.763, Recall=0.774, F1=0.768
Macro F1 = 0.712

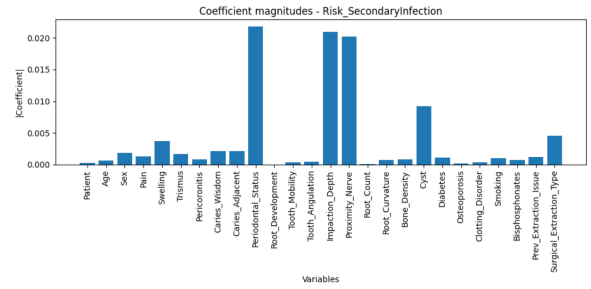
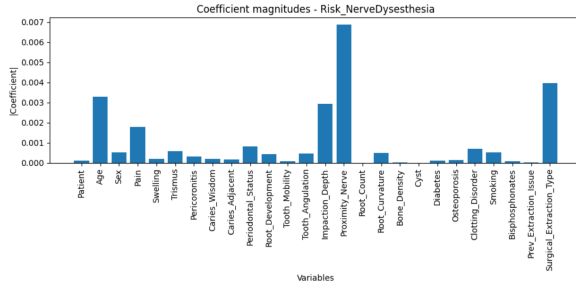
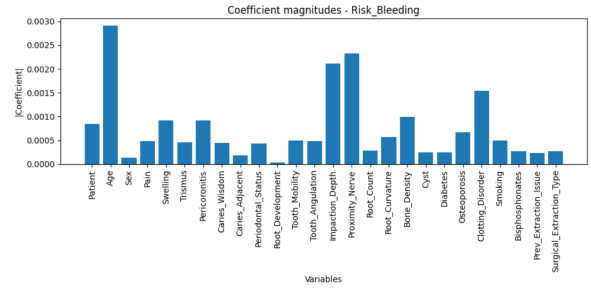
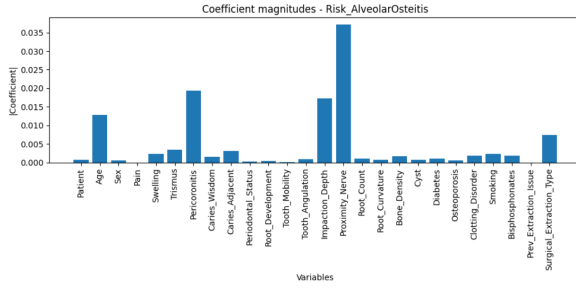
--- Risk_Bleeding ---
True column: Risk_Category_Bleeding
Pred column: Risk_Bleeding_group
Class 0: Precision=0.501, Recall=0.584, F1=0.539
Class 1: Precision=0.369, Recall=0.395, F1=0.382
Class 2: Precision=0.616, Recall=0.510, F1=0.558
Macro F1 = 0.493

=====
FINAL F1 (mean over risks) = 0.630
=====

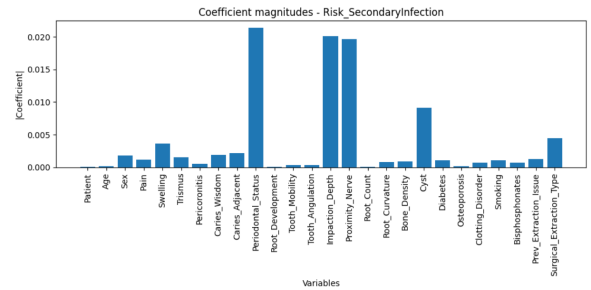
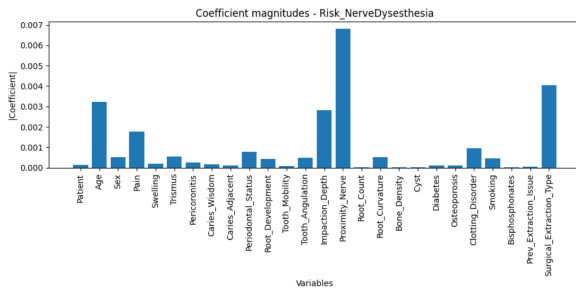
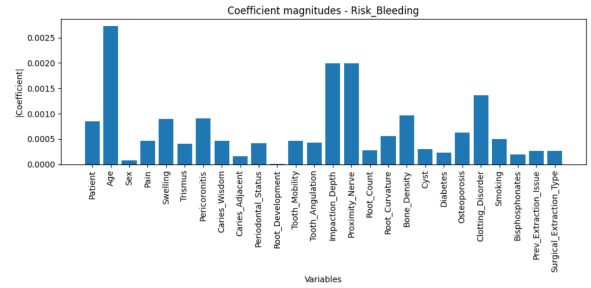
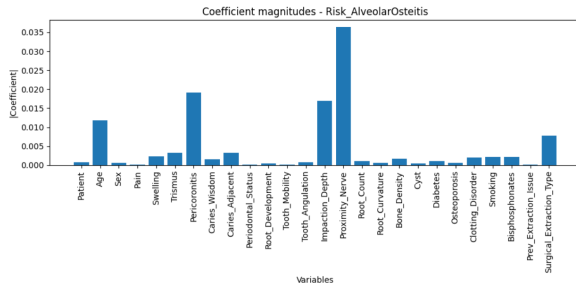
```

(b) Federated

Overall, the results are promising. The federated approach achieves performance close to the centralized setup. In addition, the achieved macro F1-score is relatively high compared to the performance observed for the MLP model (0.7204 for centralized and 0.6414 for federated). The plots below show the regression coefficients for the four centralized regression models, one for each complication type.



The coefficient plots indicate that the different complication types rely on different input features. While some variables, such as impaction depth, appear important across multiple complications, others have large coefficients only for specific risks, suggesting that each complication is influenced by distinct underlying factors. Below we have a similar plot for the federated models:



We see that for the federated setup, the coefficient magnitude are almost identical to the centralized model.

We also do the same for logistic regression; however, since we see that the results are consistently worse, or similar, we decide to drop that approach now.

We expected to get similar results from logistic regression, since when assigning the categories we are only interested in the 'order' of the values and not the actual size of the probabilities, and it would therefore be the same as for multiple regression, with only small deviances, due to logistic being an iterative approach.

Feature Selection

We now turn to feature selection. To keep the analysis manageable we only do the following steps for the complication Alveolar_osteitis, doing it for the other complications would only require running the code again for these.

Due to the computational cost of several of the feature selection techniques, we in some cases reduce the size of the dataset in order to make the experiments easier to run.

Forward and Backward selection

We perform both forward and backward feature selection using a significance level of 0.05 on the model without interaction-terms. We select the following features:

const	0.088872
Age	0.012257
Trismus	0.004064
Pericoronitis	0.020224
Caries_Adjacent	-0.002893
Impaction_Depth	0.016990
Proximity_Nerve	0.037978
Surgical_Extraction_Type	0.007296

(a) Backward

const	0.088872
Proximity_Nerve	0.037978
Pericoronitis	0.020224
Impaction_Depth	0.016990
Age	0.012257
Surgical_Extraction_Type	0.007296
Trismus	0.004064
Caries_Adjacent	-0.002893

(b) Forward

Both techniques select the same variables. The selected variables are largely consistent with what was observed in the coefficient-magnitude plots for Alveolar Osteitis, where the same features appeared to have the strongest influence on the prediction.

We are not only interested in which features on their own are important, but also which interactions. Therefore we want to do feature selection on all interaction pairs. Since this is computationally heavy we limit the dataset to only 2500 data points. We select the following features using backward and forward selection:

```
[ 'Age',
  'Age_x_Proximity_Nerve',
  'Bone_Density_x_Cyst',
  'Bone_Density_x_Prev_Extraction_Issue',
  'Caries_Adjacent_x_Prev_Extraction_Issue',
  'Cyst_x_Clotting_Disorder',
  'Diabetes_x_Surgical_Extraction_Type',
  'Impaction_Depth',
  'Impaction_Depth_x_Proximity_Nerve',
  'Pericoronitis',
  'Pericoronitis_x_Bone_Density',
  'Pericoronitis_x_Proximity_Nerve',
  'Periodontal_Status_x_Osteoporosis',
  'Proximity_Nerve',
  'Proximity_Nerve_x_Bone_Density',
  'Root_Count',
  'Root_Count_x_Cyst',
  'Root_Development',
  'Root_Development_x_Proximity_Nerve',
  'Sex_x_Cyst',
  'Sex_x_Root_Count',
  'Swelling_x_Prev_Extraction_Issue',
  'Swelling_x_Root_Count',
  'Swelling_x_Surgical_Extraction_Type',
  'Tooth_Angulation_x_Clotting_Disorder',
  'Trismus',
  'Trismus_x_Cyst',
  'Trismus_x_Proximity_Nerve']
```

(a) Backward

```
[ 'Age',
  'Age_x_Proximity_Nerve',
  'Bone_Density_x_Cyst',
  'Bone_Density_x_Prev_Extraction_Issue',
  'Caries_Adjacent_x_Prev_Extraction_Issue',
  'Caries_Wisdom_x_Osteoporosis',
  'Cyst_x_Clotting_Disorder',
  'Diabetes_x_Surgical_Extraction_Type',
  'Impaction_Depth',
  'Impaction_Depth_x_Proximity_Nerve',
  'Pericoronitis',
  'Pericoronitis_x_Bone_Density',
  'Pericoronitis_x_Proximity_Nerve',
  'Proximity_Nerve',
  'Proximity_Nerve_x_Bone_Density',
  'Root_Count',
  'Root_Count_x_Cyst',
  'Root_Development',
  'Root_Development_x_Proximity_Nerve',
  'Sex_x_Cyst',
  'Sex_x_Root_Count',
  'Swelling_x_Prev_Extraction_Issue',
  'Swelling_x_Root_Count',
  'Swelling_x_Surgical_Extraction_Type',
  'Tooth_Angulation_x_Clotting_Disorder',
  'Trismus',
  'Trismus_x_Cyst',
  'Trismus_x_Proximity_Nerve']
```

(b) Forward

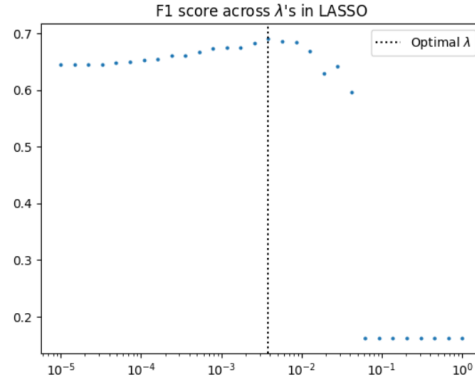
Both methods select 28 features with only a few differences. The selected features suggest that interaction effects play an important role in predicting Alveolar Osteitis. Several interaction terms are not eliminated, indicating that the effect of one variable often depends on the value of another variable.

However, forward and backward selection only work for the centralized setup and not for federated, since that would require aggregating p-values (which does not make sense). Therefore we also look into using LASSO-regression, since LASSO-regression can be adapted to the federated setup.

LASSO

We now investigate LASSO regression as a method for feature selection in both the federated and centralized setup. By adding regularization during training, LASSO can reduce the number of selected features. Note that we are now working on the full dataset.

First we do it on a centralized model, in order to have something to compare the federated setup results to. In order to do this we first need to choose a good λ -value, and we therefore do a sweep off 30 different λ - values. The plot below shows the F1-macro score vs lambda value:



We now investigate the performance of the best observed λ -value. However, as seen in the previous plot, a fairly large range of λ -values gives similar performance, indicating that the model is relatively robust to the exact choice of regularization strength. The lambda value with the best F1-score is 0.00385. With the following performance and selected features:

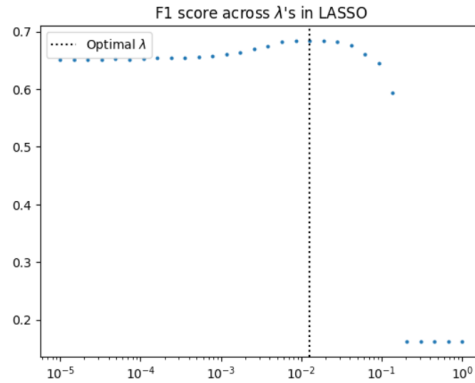
```
{'per_class': {0: {'TP': np.int64(794),
'FP': np.int64(212),
'FN': np.int64(171),
'precision': np.float64(0.7892644135188867),
'recall': np.float64(0.8227979274611399),
'f1': np.float64(0.8056823947234907)},
1: {'TP': np.int64(565),
'FP': np.int64(432),
'FN': np.int64(462),
'precision': np.float64(0.5667001003009027),
'recall': np.float64(0.5501460564751705),
'f1': np.float64(0.558300395256917)},
2: {'TP': np.int64(710),
'FP': np.int64(287),
'FN': np.int64(298),
'precision': np.float64(0.7121364092276831),
'recall': np.float64(0.7043650793650794),
'f1': np.float64(0.7082294264339153)}},
'macro_f1': np.float64(0.6907374054714411)}
```

```
['Age_x_Impaction_Depth',
'Age_x_Proximity_Nerve',
'Age_x_Surgical_Extraction_Type',
'Trismus_x_Proximity_Nerve',
'Pericoronitis_x_Impaction_Depth',
'Pericoronitis_x_Proximity_Nerve',
'Impaction_Depth_x_Proximity_Nerve',
'Impaction_Depth_x_Surgical_Extraction_Type']
```

The centralized lasso regression with lambda value 0.00385 selects 8 features and achieves a performance of 0.69, which is an improvement from 0.676, which we achieved for Alveolar osteitis without interactions. The centralized MLP-model achieves a performance of 0.818 for alveolar osteitis.

Interestingly, when optimizing for F1-score, like this, in a centralized setup we choose a lot fewer variables than in the forward/backward setup.

We then do it for the federated model. The plot below shows the sweep:



We now investigate the performance of the best observed λ -value. Again we observe, that a

fairly large range of λ -values gives similar performance, indicating that the model is relatively robust to the exact choice of regularization strength. The lambda value with the best F1-score is 0.012689. With the following performance and selected features:

```
[ 'Age',
  'Impaction_Depth',
  'Proximity_Nerve',
  'Age_x_Pericoronitis',
  'Age_x_Impaction_Depth',
  'Age_x_Proximity_Nerve',
  'Age_x_Surgical_Extraction_Type',
  'Trismus_x_Impaction_Depth',
  'Trismus_x_Proximity_Nerve',
  'Trismus_x_Root_Curvature',
  'Pericoronitis_x_Tooth_Mobility',
  'Pericoronitis_x_Impaction_Depth',
  'Pericoronitis_x_Proximity_Nerve',
  'Pericoronitis_x_Surgical_Extraction_Type',
  'Caries_Wisdom_x_Diabetes',
  'Periodontal_Status_x_Proximity_Nerve',
  'Root_Development_x_Proximity_Nerve',
  'Impaction_Depth_x_Proximity_Nerve',
  'Impaction_Depth_x_Surgical_Extraction_Type',
  'Proximity_Nerve_x_Root_Curvature',
  'Proximity_Nerve_x_Surgical_Extraction_Type']
```

```
{'per_class': {0: {'TP': np.int64(788),
  'FP': np.int64(212),
  'FN': np.int64(177),
  'precision': np.float64(0.788),
  'recall': np.float64(0.816580310880829),
  'f1': np.float64(0.8020356234096693)},
  1: {'TP': np.int64(553),
  'FP': np.int64(447),
  'FN': np.int64(474),
  'precision': np.float64(0.553),
  'recall': np.float64(0.5384615384615384),
  'f1': np.float64(0.5456339417858905)},
  2: {'TP': np.int64(707),
  'FP': np.int64(293),
  'FN': np.int64(301),
  'precision': np.float64(0.707),
  'recall': np.float64(0.7013888888888888),
  'f1': np.float64(0.704183266932271)}},
'macro_f1': np.float64(0.6839509440426103)}
```

The federated lasso regression with lambda value 0.012689 selects 21 features (so a lot more than in the centralized model) and achieves a performance of 0.68, which is an improvement from 0.66, which we achieved in the multiple regression model for Alveolar osteitis without interactions. The federated MLP-model achieves a F1-macro score of 0.7152 for alveolar osteitis. It should be noted that the λ values are selected directly based on performance on the test set. From a methods perspective, this is not fully correct, since the test set should ideally only be used for final evaluation and not for hyperparameter tuning. As a result, the reported performances should be interpreted as an upper bound on the achievable performance.

Beta Sweep Results

In this section, we investigate the performance of our MLP model under different levels of data heterogeneity in a federated learning setting. So far, we have worked on a synthetic dataset generated by Jonas, who previously worked on the project.

Our data partitioning framework distributes the data across clients using two parameters: β_L and β_Q . The parameter β_Q controls the quantity skew, i.e., how many data points each client receives. The parameter β_L controls the label skew by influencing the distribution of patients with complications across clients.

Smaller values of β_L and β_Q result in a more heterogeneous federated setup, while larger values lead to more homogeneous client datasets. This enables us to systematically analyze how varying levels of heterogeneity affect the performance of our federated MLP model.

The Work can be found in `src/analysis`.

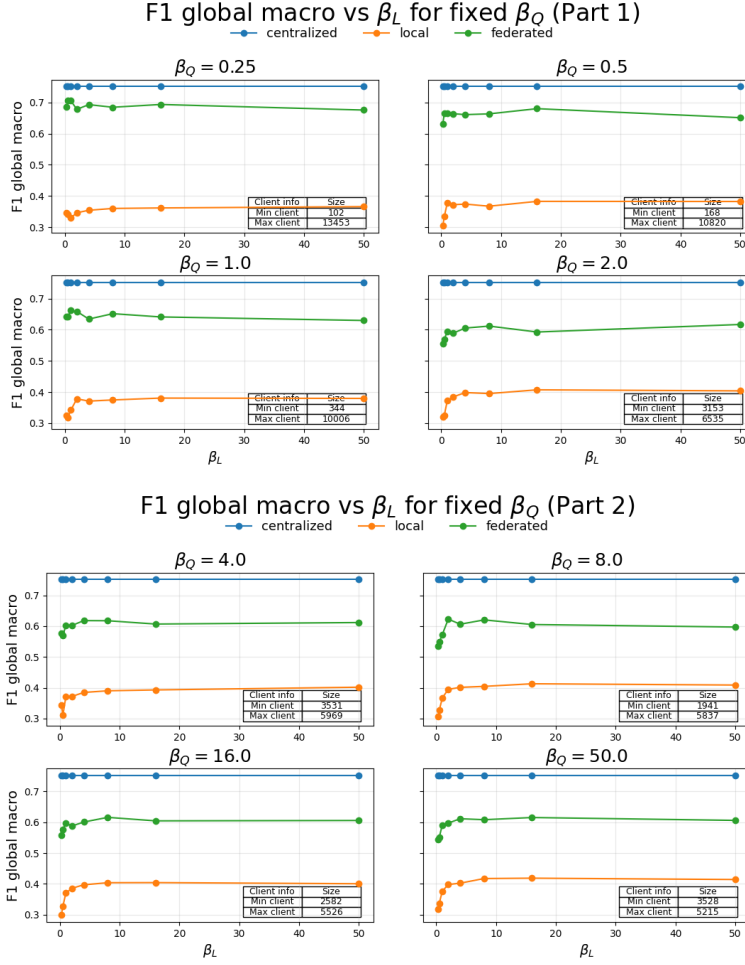
Label Skew

We run a grid search/beta sweep. Using seed 42 (and a dataset of size 50000 with 3000 test data points), with the values:

$$\beta_Q = [0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 50.0]$$

$$\beta_L = [0.25, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 50.0]$$

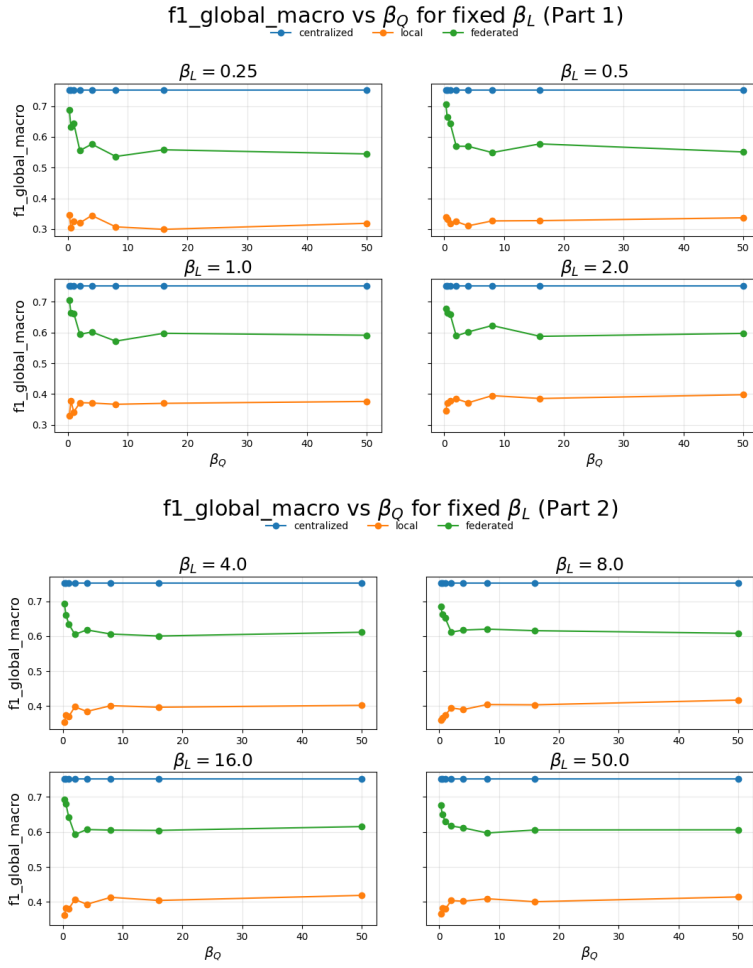
The eight plots below show the F1 macro metric (computed using global thresholds) as a function of label skew (β_L) for a fixed quantity skew (β_Q).



The results show a clear performance ranking between the three training paradigms. Centralized training consistently achieves the highest F1 score, followed by federated learning, while locally trained models perform substantially worse. This confirms that federated learning successfully leverages information across clients, significantly outperforming purely local training. Varying the label skew parameter β_L generally has a limited impact on model performance for small values of β_Q , where quantity skew is high. However, as β_Q increases and the client datasets become more balanced in size, the effect of label skew becomes more important. In particular, very low values of β_L lead to a decrease in federated performance for $\beta_Q \geq 2$. This suggests that the federated model is relatively robust to moderate levels of label heterogeneity, but extreme label skew can harm performance when the quantity distribution across clients is more homogeneous. High quantity skew (small β_Q s) results in a high performance of the federated model. This is likely due to that few/one client has most of the datapoints resulting in a model close to the centralized model.

Quantity Skew

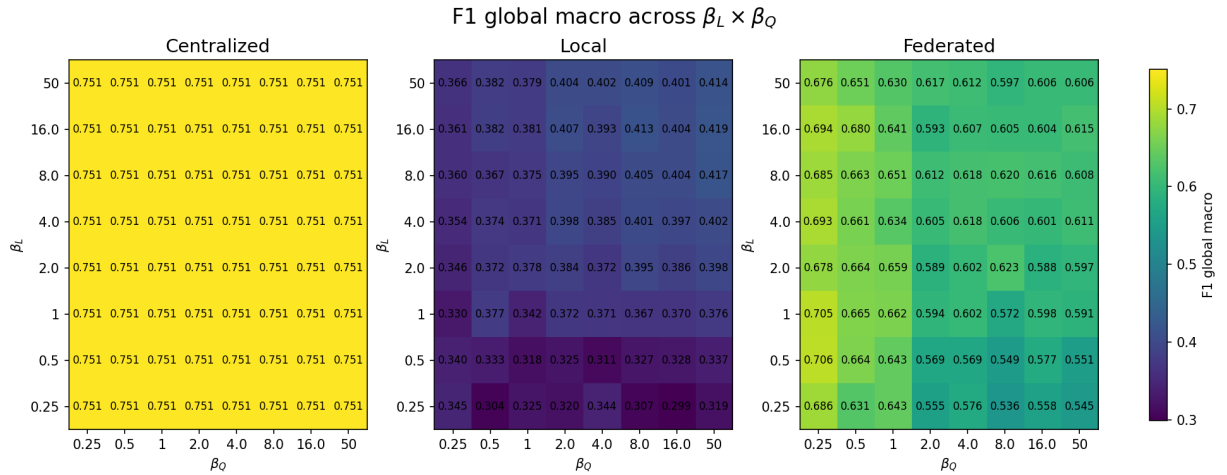
The eight plots below show the F1 macro metric (computed using global thresholds) as a function of quantity skew (β_Q) for a fixed label skew (β_L).



The plot shows the same trends as before, but now from the perspective of varying β_Q for fixed values of β_L .

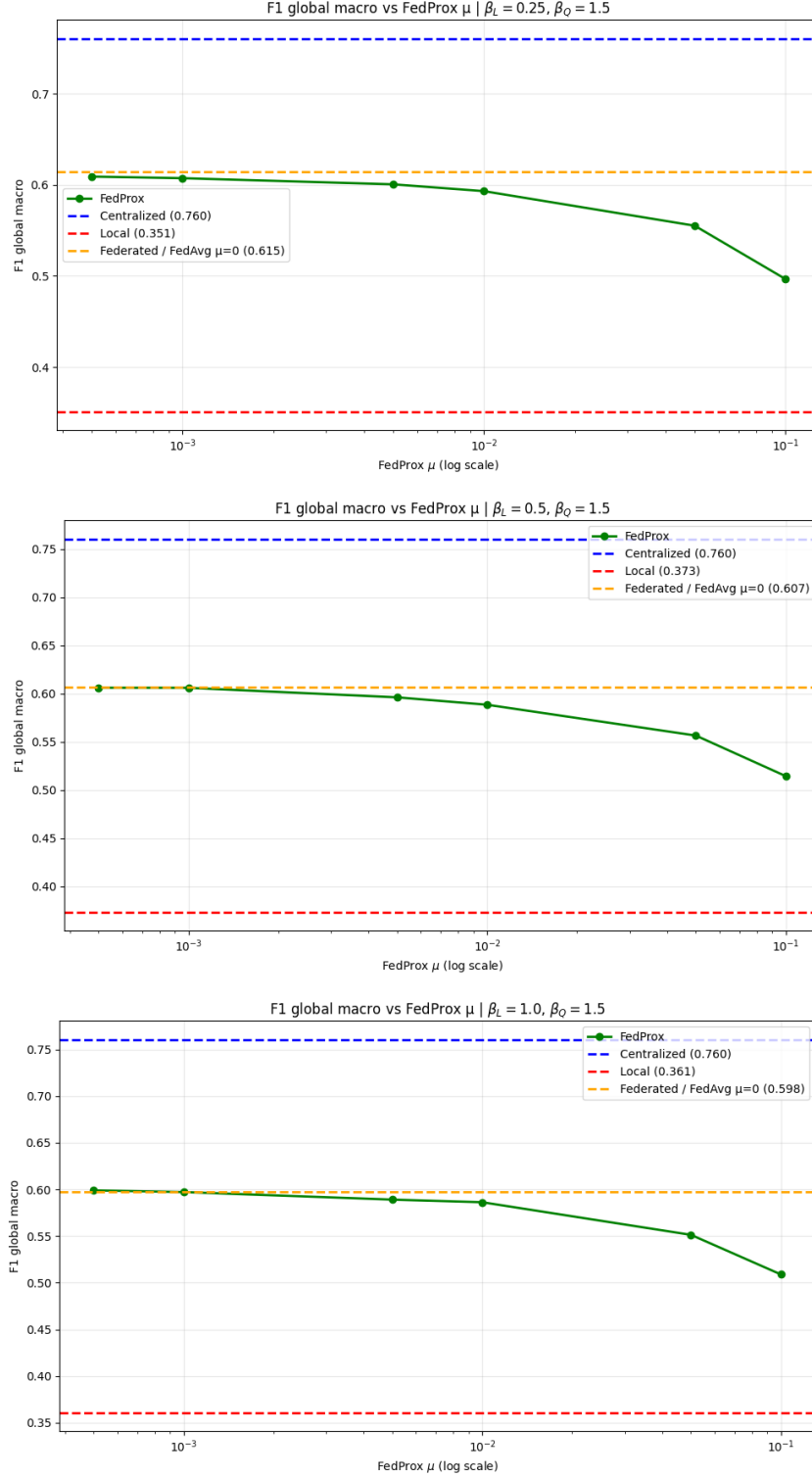
Interaction Plot

The heatmaps summarize the global macro F1 performance across combinations of β_L and β_Q for the centralized, local, and federated setups.



FedProx

Below is three plots of the FedProx model. The x-axis has the μ -value. For reference we have centralized, local and FedAvg show by dashed lines. In the three plots we have used a moderate quantity skew of $\beta_Q = 1.5$. In the three plot we vary the label skew with the values 0.25,0.5 and 1.0:



These findings indicate that the additional proximal constraint introduced by FedProx may be unnecessary in scenarios where client heterogeneity is relatively moderate.

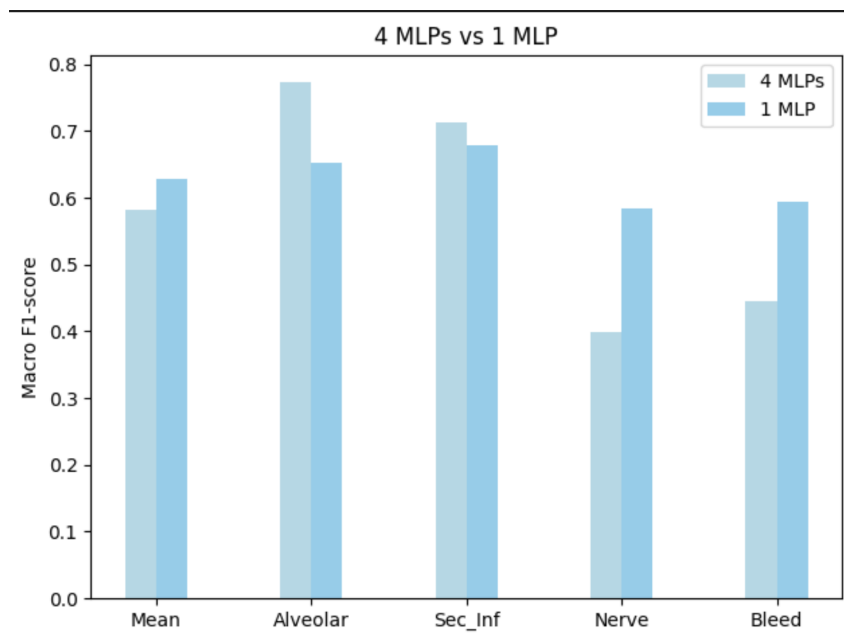
Optimizing MLP-architecture

In this section we experiment with different architectures of the MLP. The motivation for this is that we through our simple models have seen that different features in the dataset have different importance for different complications. Because of this a single MLP with only fully connected layers might not be optimal. We try to train four independent MLP's and we also try to train an MLP, that has some shared layers and then branching out into individual layers for each complication. We also try different architectures of the single MLP

The work can be found in `src/fdrp/analysis/results/MLP_arkitektur_results`.

Four independent MLP's

We train four different MLP's - one for each complication - and compute the F1-score for each complication as well as the mean F1-score.



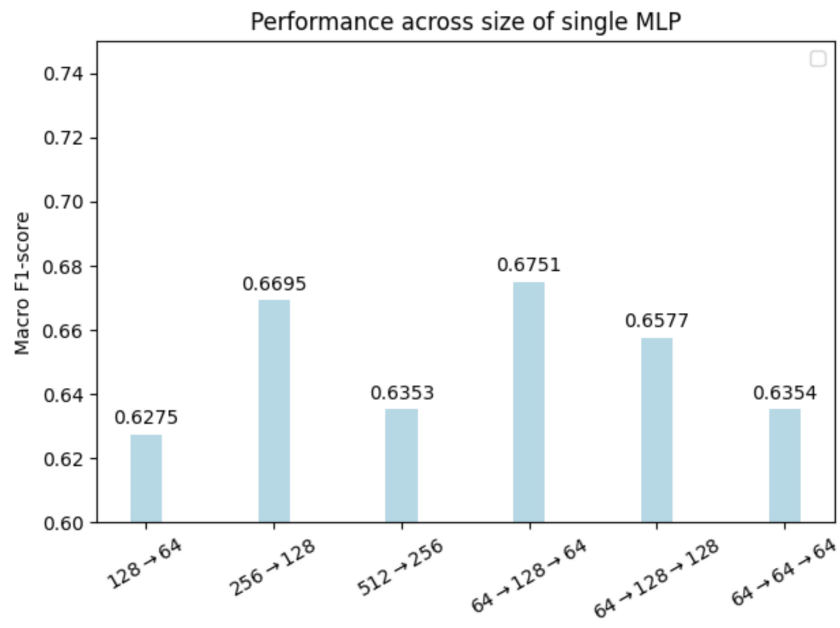
We observe above that there is greater variance in performance across complications for the four MLPs. Performance is very poor for the rare categories, “Nerve” and “Bleeding”. We hypothesize that this is because the rare complications contain too few positive cases for the model to learn the patterns properly. Therefore, it is advantageous to use a shared model that can learn general features that are relevant across complications, such as `nerve_proximity`.

From this, we conclude that four individual MLPs perform poorly and are not a suitable architecture.

Models with branching

1 model, no branching, different sizes of layers

To see if the increased performance of the branching models was just due to their size increasing, we did some experiments on the single MLP and its size and architecture. We tried with different number of hidden layers and different number of neurons in these hidden layers. Below are the results. The notation used is for example $128 \rightarrow 64$ which means a hidden layer of size 128 into a hidden layer of size 64 - giving us a total of 2 hidden layers.



Comparison of the above 3